VEKTOR

BY

INVESTPUPPY

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Per-Instrument Signal Optimisation & XGBoost Validation

How Vektor selects the optimal technical indicator for every individual holding — and why one-size-fits-all signal strategies leave alpha on the table.

For professional and institutional investors

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ABSTRACT

One-size-fits-all signal selection is where most systematic portfolios leave alpha on the table. This paper describes exactly how Vektor recovers it — per instrument, not on average. Standard quantitative portfolio strategies apply a single technical indicator across all holdings — a compromise that optimises for the average at the expense of the individual. This paper describes Vektor's per-instrument signal optimisation methodology: a full grid search across six technical indicators for each holding, Sharpe-ranked parameter selection, and a two-stage XGBoost validation layer that filters live signals before execution. The result is a portfolio where every position is driven by the indicator that actually worked for that specific instrument — not the one that performed best on average across the universe.

KEY TAKEAWAYS

— A single technical indicator applied across a universe is a known source of lost alpha. Per-instrument selection recovers it.

— Grid search across six indicators with full parameter optimisation produces the highest-Sharpe configuration for each individual stock.

— XGBoost validation adds a second layer of signal verification, filtering out low-confidence signals before they reach execution.

— The winning indicator, its optimised parameters, and its backtest Sharpe ratio are stored in the strategy record — fully transparent and auditable.

1. THE PROBLEM WITH UNIVERSAL SIGNALS

Why a single indicator applied across a portfolio is a structural source of lost alpha.

Technical indicators are designed to detect specific price behaviours — momentum, mean reversion, volatility breakouts, overbought/oversold conditions. Different instruments exhibit different dominant price behaviours at different points in time. A financial services stock in Singapore may trend strongly and respond well to a simple moving average crossover. A smaller industrial stock in the same universe may exhibit high volatility and respond better to a Bollinger Band breakout signal. Applying the same indicator to both instruments does not produce the best outcome for either — it produces a mediocre outcome for both.

This is not a theoretical concern. In a ten-stock portfolio, if three holdings are generating signals from a sub-optimal indicator, the portfolio's systematic edge is materially reduced. Across a larger mandate, the effect compounds. The solution is not a better universal indicator — it is per-instrument selection.

> The six indicators evaluated by Vektor — SMA, EMA, MACD, Bollinger Bands, RSI, and Stochastic Oscillator — were selected to cover the primary categories of price behaviour: trend-following, momentum, volatility, and mean reversion. Together they represent the full spectrum of systematic signal types available to a discretionary portfolio manager.

2. THE SIX INDICATORS

What each indicator measures, how it is parameterised, and what price behaviour it is designed to detect.

Simple Moving Average (SMA)

Crossover of a short-term and long-term moving average. Generates a buy signal when the short average crosses above the long, and a sell signal on the reverse. Best suited to instruments with persistent trending behaviour. Optimised parameters: Short period 10–30 days · Long period 40–80 days.

Exponential Moving Average (EMA)

Identical structure to SMA but applies exponentially greater weight to recent prices, making it more responsive to current market conditions. Better suited to instruments with trend reversals or regime changes. Optimised parameters: Short period 10–30 days · Long period 40–80 days.

MACD

Momentum indicator using the convergence and divergence of two EMAs. Generates signals when the MACD line crosses its signal line. Effective for instruments with cyclical momentum patterns. Optimised parameters: Short EMA 5–20 · Long EMA 20–50 · Signal 5–15.

Bollinger Bands (BB)

Volatility-based bands set at standard deviations around a moving average. Generates signals when price breaks out of or returns to the bands. Suited to instruments with periodic volatility expansions. Optimised parameters: SMA period 10–30 · Standard deviations 1–3.

Relative Strength Index (RSI)

Momentum oscillator measuring the speed and magnitude of price movements. Generates signals at overbought and oversold thresholds. Suited to instruments with clear mean-reverting tendencies. Optimised parameters: Period 10–20 · Upper threshold 60–80 · Lower threshold 20–40.

Stochastic Oscillator (SO)

Compares the closing price to the price range over a look-back period. Generates signals at extremes of the range. Similar in intent to RSI but with different sensitivity characteristics. Optimised parameters: Period 10–20 · %D smoothing 2–5.

3. THE GRID SEARCH METHODOLOGY

How Vektor searches the full parameter space for each indicator and each instrument.

For each instrument in the portfolio and each of the six indicators, Vektor executes a full grid search across the indicator's parameter space. The grid search tests every combination of parameter values within the defined ranges. Each combination is backtested against three years of daily price data and the annualised Sharpe ratio is calculated. The Sharpe ratio is selected because it simultaneously captures both the return and the risk of the signal's performance.

Step 1 — Define parameter grid: the full range of valid parameter combinations is enumerated for each indicator.

Step 2 — Backtest each configuration: applied to three years of daily OHLCV price data, generating buy/sell signals and performance statistics.

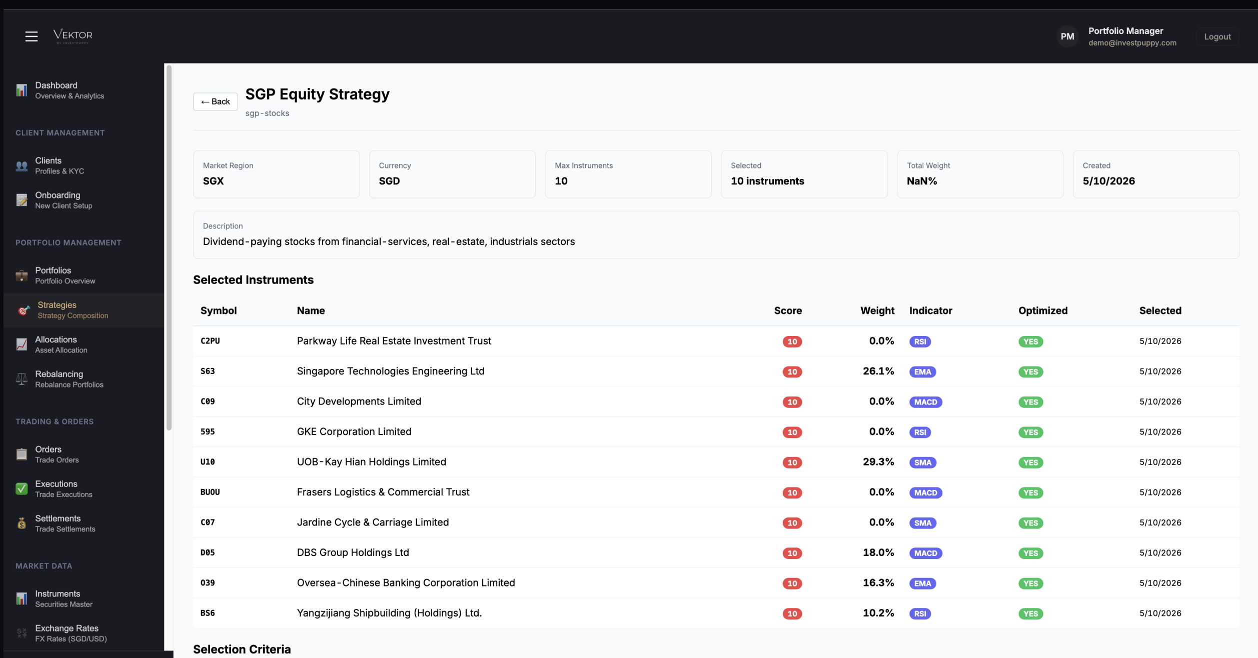
Step 3 — Calculate annualised Sharpe ratio for each configuration.

Step 4 — Select highest-Sharpe configuration as the winner for that instrument.

Step 5 — Repeat for all six indicators; the indicator with the highest Sharpe across all six is assigned.

Step 6 — Save to strategy record: indicator name, optimised parameters, and backtest Sharpe ratio — fully visible to the portfolio manager before any trade approval.

PLATFORM INTERFACE



Symbol	Name	Score	Weight	Indicator	Optimized	Selected
C2PU	Parkway Life Real Estate Investment Trust	10	0.0%	RSI	YES	5/10/2026
S63	Singapore Technologies Engineering Ltd	10	26.1%	EMA	YES	5/10/2026
C09	City Developments Limited	10	0.0%	MACD	YES	5/10/2026
595	GKE Corporation Limited	10	0.0%	RSI	YES	5/10/2026
U10	UOB-Kay Hian Holdings Limited	10	29.3%	SMA	YES	5/10/2026
BUOU	Frasers Logistics & Commercial Trust	10	0.0%	MACD	YES	5/10/2026
C07	Jardine Cycle & Carriage Limited	10	0.0%	SMA	YES	5/10/2026
D05	DBS Group Holdings Ltd	10	18.0%	MACD	YES	5/10/2026
O39	Oversea-Chinese Banking Corporation Limited	10	16.3%	EMA	YES	5/10/2026
BS6	Yangzijiang Shipbuilding (Holdings) Ltd.	10	10.2%	RSI	YES	5/10/2026

SGP Equity Strategy — admin portal. Each instrument shows its assigned indicator (BB, MACD, SMA, EMA) alongside its allocation weight and optimisation status.

4. XGBOOST SIGNAL VALIDATION

Why a second validation layer is necessary — and how XGBoost provides it.

The grid search selects the optimal historical configuration for each instrument. But a signal that performed well in backtest is not guaranteed to be valid at the moment of execution. Market regimes change. A historically strong MACD signal may be generated during a period of low liquidity, unusual volatility, or data quality issues. The XGBoost layer addresses this by providing real-time validation of every live signal before it reaches the execution layer.

XGBoost (Extreme Gradient Boosting) is a machine learning classification model trained on historical signal outcomes — specifically, whether a signal generated by the selected indicator in similar market conditions historically produced a positive outcome. The model outputs a confidence score for each live signal. Signals below the confidence threshold are filtered out before execution, preventing the system from acting on historically weak signals even if the indicator technically generated them.

> Two layers of rigour: the grid search optimises the signal for the instrument's historical behaviour. XGBoost validates the signal against current market conditions. Neither layer alone is sufficient — together they provide the systematic edge that neither a single indicator nor a pure ML model would achieve independently.

5. TRANSPARENCY AND AUDITABILITY

Every signal selection decision is recorded, visible, and explainable.

A portfolio manager using Vektor can examine, for every holding in every strategy: which indicator was selected and why, what parameters were optimised, what backtest Sharpe ratio justified the selection, and whether the XGBoost layer validated the most recent live signal. Nothing is a black box. If a position cannot be explained, the system has failed — and Vektor is built so that failure cannot happen silently.

6. CONCLUSION

Per-instrument signal optimisation is not a marginal improvement over universal signal strategies — it is a structural change in how systematic signals are applied.

By searching the full parameter space for each of six indicators across every instrument, and validating every live signal through an XGBoost layer, Vektor ensures that each position in the portfolio is driven by the signal that actually works for that specific instrument — not the signal that works on average. In a ten-stock portfolio, the difference between an optimised per-instrument signal and a universal signal is the difference between a systematic edge and a systematic compromise.

> *Each position in the portfolio is driven by the indicator that actually works for that instrument — not the one that performs best on average. That is the systematic edge per-instrument optimisation produces: not a marginal improvement, but a structural change in how every signal in the portfolio is selected.*

This paper describes Step 04 of the Vektor eleven-step workflow. For the full portfolio construction methodology, see WP-01: Quantitative Portfolio Construction. For the commercial case, see Why Vektor. All documents available at investpuppy.com.

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